Rain-Robust Pothole Detection

How much does rain hurt a pothole detector, and what helps?

CSE36501 Deep Learning — Project Proposal

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Motivation

- Potholes cause accidents, tyre and suspension damage, and costly repairs.
- Vision-based road monitoring and driver assistance increasingly detect them automatically from vehicle cameras.
- Rain is exactly when potholes are most dangerous — and hardest to see:
  - water fills the hole and hides its depth and edges,
  - wet asphalt reflects light; rain streaks and darkness lower contrast.
- Public pothole detectors are trained almost entirely on *clear-weather* images.

A detector that fails in the rain fails when it is needed most.

Problem statement

Task. Single-class object detection (bounding boxes) of potholes in vehicle- or pedestrian-view road images.

Setting. Train on clear-weather road images; test on clear, synthetic-rain and real-rain images.

RQ1. How much does a pothole detector trained only on clear-weather images degrade under rain?

RQ2. Can synthetic rain overlays serve as a proxy for real rainy conditions?

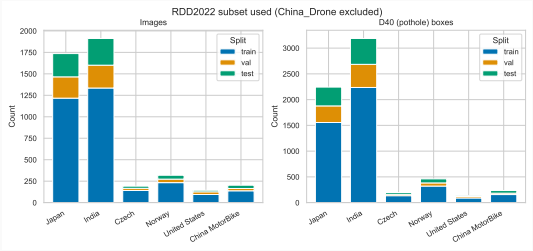
RQ3. Which strategy improves rain robustness most: (a) synthetic-rain augmentation, (b) architectural change (attention module), or (c) test-time deraining?

Datasets

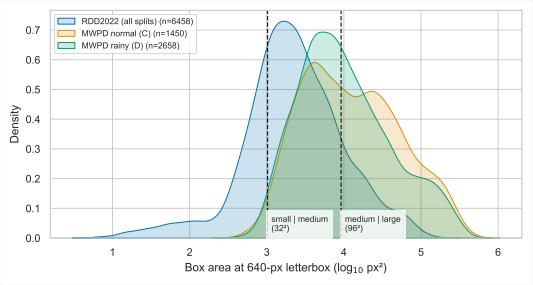
	RDD2022 (Arya et al., 2024)	MWPD (Parvin et al., 2025)
Role	clear-weather train / val / test	external test: real normal vs. rain
Source	vehicle-mounted cameras & phones, 6 countries	mixed web, news and phone images
Size	47,420 images, 4 damage types; we keep **D40 = pothole** (3,610 positive images, 6,458 boxes)	1,357 images, 640×640
Labels, license	Pascal VOC XML; CC BY 4.0	YOLO txt, 1 class; CC BY 4.0
Conditions	clear daylight	normal / rainy; day / twilight / night

- **Why these:** RDD2022 is the largest public multi-country road-damage set; MWPD is, to our knowledge, the only public pothole set with real rain.
- China_Drone excluded (top-down drone view $\neq$ vehicle view).
- MWPD ships without weather labels $\Rightarrow$ every image was tagged normal/rainy (pre-tag + manual review).

Dataset analysis



Only 10% of annotated RDD images contain a pothole; Japan + India hold 81% of them.

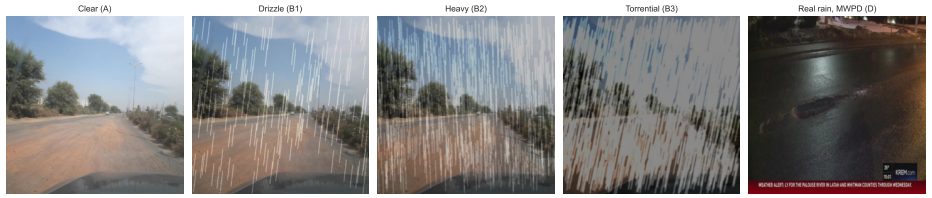


MWPD potholes are much larger (close-up photos): 26% of RDD boxes are COCO-small vs. 1% in MWPD.

Leakage-safe split: RDD frames are grouped into blocks of 50 consecutive frames per country; whole blocks go to train/val/test (70/15/15). Negatives capped at 20% per split.

Test-set design

ID	Composition	Images	Purpose
A	RDD2022 own test split (clear)	708	baseline
B1/B2/B3	*same images* as A + synthetic drizzle / heavy / torrential	708 each	pure rain effect
C	MWPD normal weather	681	domain gap
D	MWPD rainy	676	real rain effect



A→B: synthetic rain only A→C: domain (country, camera, size) C→D: real rain, same source

(A→B) vs. (C→D): is synthetic rain a valid proxy? (RQ2)

Plan & timeline

Step	Content	Status
1–2	Problem, datasets, splits, test sets A–D	done (this talk)
3	Models: YOLOv8n/s, RT-DETR-L, YOLOv8s + CBAM (ours); strategies: clean, rain-aug, deraining, real-rain upper bound	next
4	Evaluation: mAP50(-95), recall, relative drop, size-wise AP, FPS; 3 seeds	
5	Analysis, failure cases, Grad-CAM; report & final video	

Compute: one RTX 3090 (24 GB). Everything is scripted (`make data`, `make train`, ...) and seeded for reproducibility.